guideline summaries

Optimum Imaging Strategies for Advanced Prostate Cancer: ASCO Guideline Summary

Edouard J. Trabulsi, MD[1]; R. Bryan Rumble, MSc[2]; and H. Alberto Vargas, MD[3]

The purpose of this clinical practice guideline summary is to provide referring and imaging clinicians (including medical oncologists, radiation oncologists, urologists, radiologists, nuclear medicine physicians, and molecular imagers), other health care practitioners, patients, and caregivers with recommendations and future directions regarding optimum imaging for patients with advanced prostate cancer based on the best available evidence. The fluid and rapidly evolving nature of the topic is acknowledged, and although regulatory approvals of some of the techniques presented are currently limited, this guideline summary is intended to preemptively address the ongoing barrage of studies that will most certainly transform the landscape for the management of patients with advanced prostate cancer. The term *advanced prostate cancer* encompasses a wide swath of patients with different disease states and clinicopathologic factors, including men with localized prostate cancer at initial diagnosis with a high or very high risk of metastasis (as defined recently in an American Urological Association/American Society of Radiation Oncology/Society of Urologic Oncology Guideline)[1]; men who have been treated and subsequently present with clinical, biochemical, or radiographic evidence of disease progression; and men with known metastatic disease either at initial presentation or after one or more lines of treatment. This guideline summary examines the optimal use of imaging for men in each of these disease states.

Prostate cancer is the most common nondermatologic cancer in men. In 2019, it was estimated that there would be 174,650 new cases in the United States, and, in spite of advances in diagnosis and treatment, an estimated 31,620 deaths would occur.[2] In addition to its prevalence, prostate cancer poses unique challenges, including a distinct clinical disease state characterized by an elevated serum prostate specific antigen (PSA) consistent with recurrent disease without findings of metastases on historically conventional imaging studies, and difficulty in monitoring patients with metastatic bone disease due to the poor test characteristics of conventional bone imaging. For these reasons, coupled with increasing evidence for local salvage therapy or metastasis-directed therapy or increasingly effective systemic therapies that have been shown to be beneficial ever earlier in the natural history of the disease, there is great interest in identifying better imaging strategies to inform the optimum management for patients with advanced prostate cancer.

The predilection for prostate cancer to metastasize to bone and lymph nodes requires both bone and soft tissue imaging techniques to assess for staging and to monitor for response to therapy and progression of disease. Conventional standard imaging modalities include ^{99m}Tc-labeled methylene diphosphonate (MDP) bone scan and computerized tomography (CT) scan or magnetic resonance imaging (MRI). The relatively poor specificity of ^{99m}Tc-MDP relates to the fact that MDP adsorbs onto the crystalline hydroxyapatite mineral of bone and is not prostate cancer specific, thus providing only an indirect measure of tumor activity based on osteoblastic activity in the tumor microenvironment. In addition, the interpretation of a bone scan is subjective, relying on manual assessments of lesion number, size, and intensity. Such subjective assessments are problematic in the setting of clinical trials and lead to difficulty in accurately interpreting treatment effects. Patients treated with effective therapy can have paradoxically worsening scans in the face of response to treatment.[3] Standard criteria have been developed to standardize bone scan interpretation and to distinguish response from progression.[4,5] These semiquantitative criteria have been shown to be feasible, have demonstrated in prospective studies to correlate with survival, and have been recognized as clinically relevant to warrant their use by regulatory authorities for drug approval.[6-8] In addition, quantitative tools for the assessment of bone scan data have been developed, including the automated bone scan index, which represents the total tumor burden as the fraction of the total skeleton weight.[9-11] These quantitative tools improve bone scan interpretation but are still subject to the relatively poor specificity of ^{99m}Tc-MDP in patients with metastatic bone disease, the inability to detect nodal and visceral metastatic disease, and the poor ability of ^{99m}Tc-MDP and CT or MRI to detect metastatic disease in patients with early biochemical recurrence.

In the same way that molecularly targeted therapies have transformed decision making of many diseases, advances in nuclear medicine and molecular imaging

Author affiliations and support information (if applicable) appear at the end of this article.

Accepted on November 27, 2019 and published at ascopubs.org/journal/op on January 16, 2020: DOI https://doi.org/10.1200/JOP.19.00699

Endorsed by the Society of Abdominal Radiology, American College of Radiology, Society of Nuclear Medicine and Molecular Imaging, American Urological Association, American Society for Radiation Oncology, and the Society of Urologic Oncology

Reprint Requests: 2318 Mill Road, Suite 800, Alexandria, VA 22314; guidelines@asco.org

THE BOTTOM LINE

Optimum Imaging Strategies for Advanced Prostate Cancer: ASCO Guideline Summary

Guideline Question

What are the optimum imaging options that should be offered to patients with advanced prostate cancer?

Target Population

Men with advanced prostate cancer, including newly diagnosed clinical high-risk disease, suspected or confirmed metastatic disease, recurrent disease, or progressive disease while under treatment.

Target Audience

Medical oncologists, radiation oncologists, urologists, radiologists, nuclear medicine and molecular imaging physicians, other health care practitioners, such as nurses and social workers, patients, and caregivers.

Methods

An Expert Panel was convened to develop clinical practice guideline recommendations based on a systematic review of the medical literature.

Definitions

- Conventional imaging: computed tomography (CT), bone scan, and/or prostate magnetic resonance imaging (MRI)
- Next-generation imaging (NGI): positron emission tomography (PET), PET/CT, PET/MRI, whole-body MRI
- Advanced prostate cancer: disease states/clinical scenarios described
- Biochemical recurrence: defined as detectable prostate-specific antigen (PSA) with subsequent rise after radical prostatectomy, or rise of 2 above nadir PSA achieved after radiotherapy (Phoenix criteria)[15] high risk without evidence of disease locally or distantly on conventional imaging where the definition of undetectable PSA is dependent on the assay used and may change over time as more sensitive assays become available; in general, a PSA value < 0.2 ng/mL has been considered undetectable,[1] while lower values (PSA ≤ 0.01 ng/mL) have been advocated when clinically available.[16]

Recommendations

Recommendation 1. Imaging is recommended for all patients with advanced prostate cancer. See the recommendation under Clinical Question 4 for specific details according to clinical scenario (Type of recommendation: evidence based, benefits outweigh harms; Evidence quality: moderate; Strength of recommendation: strong).

Recommendation 2. One or more of the following imaging modalities should be used for patients with advanced prostate cancer: conventional imaging (defined as CT, bone scan, and/or prostate MRI), and/or NGI (PET, PET/CT, PET/MRI, whole-body MRI), according to clinical scenario (Type of recommendation: evidence based, benefits outweigh harms; Evidence quality: moderate; Strength of recommendation: strong).

Recommendation 3. It is recommended when choosing an imaging modality that disease states and clinical scenarios as described are taken into consideration, as the imaging modality may guide treatment or change clinical treatment decisions (Type of recommendation: evidence based, benefits outweigh harms; Evidence quality: moderate; Strength of recommendation: strong).

Newly Diagnosed Clinically High-Risk/Very High Risk Localized Prostate Cancer

Conventional Imaging Negative

Recommendation 4.1. When conventional imaging (defined as CT, bone scan, and/or prostate MRI) is negative in patients with a high risk of metastatic disease, NGI (defined as PET, PET/CT, PET/MRI, whole-body MRI) may add clinical benefit, although prospective data are limited (Type: informal consensus, benefits/harm ratio uncertain; Evidence quality: weak; Strength of recommendation: moderate).

Conventional Imaging Suspicious/Equivocal

Recommendation 4.2. When conventional imaging is suspicious or equivocal, NGI may be offered to patients for clarification of equivocal findings or detection of additional sites of disease that could potentially alter management, although prospective data are limited (Type of recommendation: informal consensus, benefits/harm ratio uncertain; Evidence quality: weak; Strength of recommendation: moderate).

(continued on following page)

THE BOTTOM LINE (CONTINUED)

Rising PSA After Prostatectomy and Negative Conventional Imaging (either initial PSA undetectable with subsequent rise or PSA never nadirs to undetectable)

Recommendation 4.3. Both disease states are indicative of potentially undetected, residual local, locoregional, or micrometastatic disease, and imaging options are not distinct or different between these scenarios. The goal of therapy and the potential use of salvage local therapies in these scenarios should guide the choice of imaging. For men who are not candidates or are unwilling to receive salvage local or regional therapy, additional NGI should not be offered (Type: informal consensus, benefits/harms ratio uncertain; Evidence quality: low; Strength of recommendation: moderate).

Recommendation 4.4. For men for whom salvage radiotherapy is contemplated, NGI should be offered (PSMA imaging [where available], ^{11}C choline or ^{18}F fluciclovine PET/CT or PET/MRI, whole-body MRI and/or ^{18}F NaF PET/CT) as they have superior disease detection performance characteristics and may alter patient management (Type of recommendation: evidence based, benefits outweigh harms; Evidence quality: high; Strength of recommendation: strong).

Rising PSA After Radiotherapy and Negative Conventional Imaging

Recommendation 4.5. For men in whom salvage local or regional therapy is not planned or is inappropriate, there is little evidence that NGI will alter treatment or prognosis. The role of NGI in this scenario is unclear and should not be offered, except in the context of an institutional review board–approved clinical trial (Type of recommendation: informal consensus, benefits/harms ratio uncertain; Evidence quality: intermediate; Strength of recommendation: moderate).

Recommendation 4.6. For men for whom salvage local or regional therapy (eg, salvage prostatectomy, salvage ablative therapy, or salvage lymphadenectomy) is contemplated, there is evidence supporting NGI for detection of local and/or distant sites of disease. Findings on NGI could guide management in this setting (eg, salvage local, systemic or targeted treatment of metastatic disease, combined local and metastatic therapy). PSMA imaging (where available), ^{11}C choline or ^{18}F fluciclovine PET/CT or PET/MRI, whole-body MRI and/or ^{18}F NaF PET/CT have superior disease detection performance characteristics compared with conventional imaging and alter patient management, although data are limited (Type of recommendation: evidence based, benefits outweigh harms; Evidence quality: intermediate; Strength of recommendation: moderate).

Metastatic Prostate Cancer at Initial Diagnosis or After Initial Treatment, Hormone Sensitive

Recommendation 4.7. In the initial evaluation of men presenting with hormone-sensitive disease with demonstrable metastatic disease on conventional imaging, there is a potential role for NGI to clarify the burden of disease and potentially shift the treatment intent from multimodality management of oligometastatic disease to systemic anticancer therapy alone or in combination with targeted therapy for palliative purposes, but prospective data are limited (Type of recommendation: informal consensus, benefits/harms ratio uncertain; Evidence quality: intermediate; Strength of recommendation: moderate).

Nonmetastatic CRPC

Recommendation 4.8. For men with nonmetastatic castration-resistant prostate cancer (CRPC), NGI can be offered only if a change in the clinical care is contemplated. Assuming patients have received or are ineligible for local salvage treatment options, NGI may clarify the presence or absence of metastatic disease, but the data on detection capabilities of NGI in this setting and impact on management are limited (Type of recommendation: consensus, benefits/harms ratio uncertain; Evidence quality: weak; Strength of recommendation: moderate).

Metastatic CRPC

PSA Progression

Recommendation 4.9. As recommended by the Prostate Cancer Working Group 3 consensus statements, PSA progression alone for men on treatment of metastatic CRPC should not be the sole reason to change therapy. Conventional imaging can be used for initial evaluation of PSA progression and should be continued to facilitate changes/comparisons and serially to assess for development of radiographic progression (Type: informal consensus, benefits/harms ratio uncertain; Evidence quality: intermediate; Strength of recommendation: strong).

Recommendation 4.10. The use of NGI in this cohort is unclear, with a paucity of prospective data. When a change in clinical care is contemplated, in an individualized manner, and there is a high clinical suspicion of subclinical metastasis despite negative conventional imaging, the use of NGI could be contemplated, especially in the setting of a clinical trial (Type of recommendation: informal consensus, benefits/harms ratio uncertain; Evidence quality: insufficient; Strength of recommendation: weak).

(continued on following page)

THE BOTTOM LINE (CONTINUED)

Radiographic Progression on Conventional Imaging

Recommendation 4.11. In men with metastatic CRPC with clear evidence of radiographic progression on conventional imaging while on systemic therapy, NGI should not be routinely offered. NGI may play a role if performed at baseline to facilitate comparison of imaging findings/extent of progression of disease (Type of recommendation: consensus, benefits/harms ratio uncertain; Evidence quality: insufficient; Strength of recommendation: moderate).

See Figure 1 for the imaging algorithm for high/very high-risk disease at initial presentation (per National Comprehensive Cancer Network). See Figure 2 for the imaging algorithm for patients with rising PSA after local treatment.

Additional Resources

More information, including a Data Supplement with additional evidence tables, slide sets, and clinical tools and resources, is available at www.asco.org/genitourinary-cancer-guidelines. The Methodology Manual (available at www.asco.org/guideline-methodology) provides additional information about the methods used to develop this guideline. Patient information is available at www.cancer.net.

ASCO believes that cancer clinical trials are vital to inform medical decisions and improve cancer care, and that all patients should have the opportunity to participate.

are poised to reinvent the way in which we diagnose, stage, and monitor response to therapy in patients with prostate cancer. These next-generation imaging (NGI)

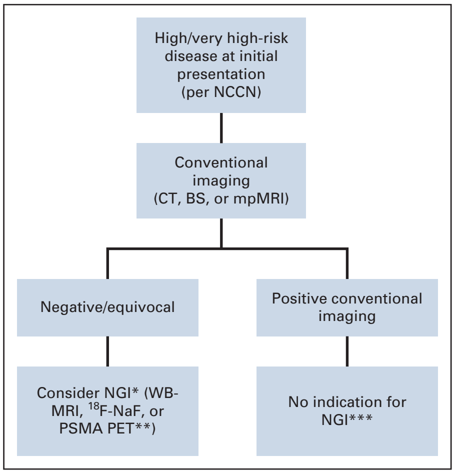

FIG 1. Imaging algorithm for high/very high-risk disease at initial presentation (per National Comprehensive Cancer Network [NCCN]). (*) Suspicious findings on NGI would influence treatment decisions in patients with advanced prostate cancer and negative conventional imaging, opening the scope for multimodality treatment of primary and oligometastatic disease or systemic therapy for more extensive metastatic states, although prospective data are limited. (**) There is enthusiasm for the potential added value of PSMA PET/CT and PET/MRI for the assessment of the local and metastatic extent of prostate cancer in this context, although PSMA imaging is not currently FDA approved and should thus be only performed as part of a clinical trial or other controlled research setting. (***) NGI could offer clinical benefit in this scenario by redefining the true extent of disease and shifting treatment decisions accordingly, although prospective data in this context are limited. BS, bone scintigraphy; CT, computed tomography; mpMRI, multiparametric magnetic resonance imaging; NGI, next-generation imaging; PET, positron emission tomography; PSMA, prostate-specific membrane antigen; WB, whole body.

modalities promise improved diagnostic accuracy for staging prostate cancer, especially at lower tumor burdens. New radiopharmaceuticals coupled to prostate cancer–specific targets, such as prostate-specific membrane antigen (PSMA; defined in this guideline summary as external domain PSMA binding ligands labeled with the positron emission tomography [PET] emitters ^{68}Ga and ^{18}F, exclusive of the PSMA antibody capromab pendetide that binds the internal domain of PSMA). These have demonstrated improvements in the sensitivity for the detection of metastatic disease and the monitoring of a treatment effect in patients with advanced prostate cancer compared with conventional imaging. The recent US Food and Drug Administration approvals of ^{11}C choline and ^{18}F fluciclovine for PET imaging in men with a suspected prostate cancer recurrence based on an elevated PSA following prior treatment represents watershed moments in the development of novel imaging tools for advanced prostate cancer.[12]

While patients and clinicians may find additional imaging, especially NGI, attractive for advanced prostate cancer in a variety of clinical states, there are several disadvantages associated with its use. These NGI modalities are costly and are not routinely covered by all third-party payors in the United States. Their routine use could significantly increase overall expenditure for prostate cancer care. While the NGI modalities have excellent sensitivity to detect low-burden disease, false-positive results may lead to incorrect patient management in some patients. Moreover, the presence of NGI-detected lesions may trigger additional procedures, imaging modalities, and invasive biopsies, which carry risks and additional cost, in addition to the mental burden and uncertainty for patients.

Another important consideration is concern for the Will Rogers phenomenon, which may ensue when patients who are classified as having nonmetastatic disease with

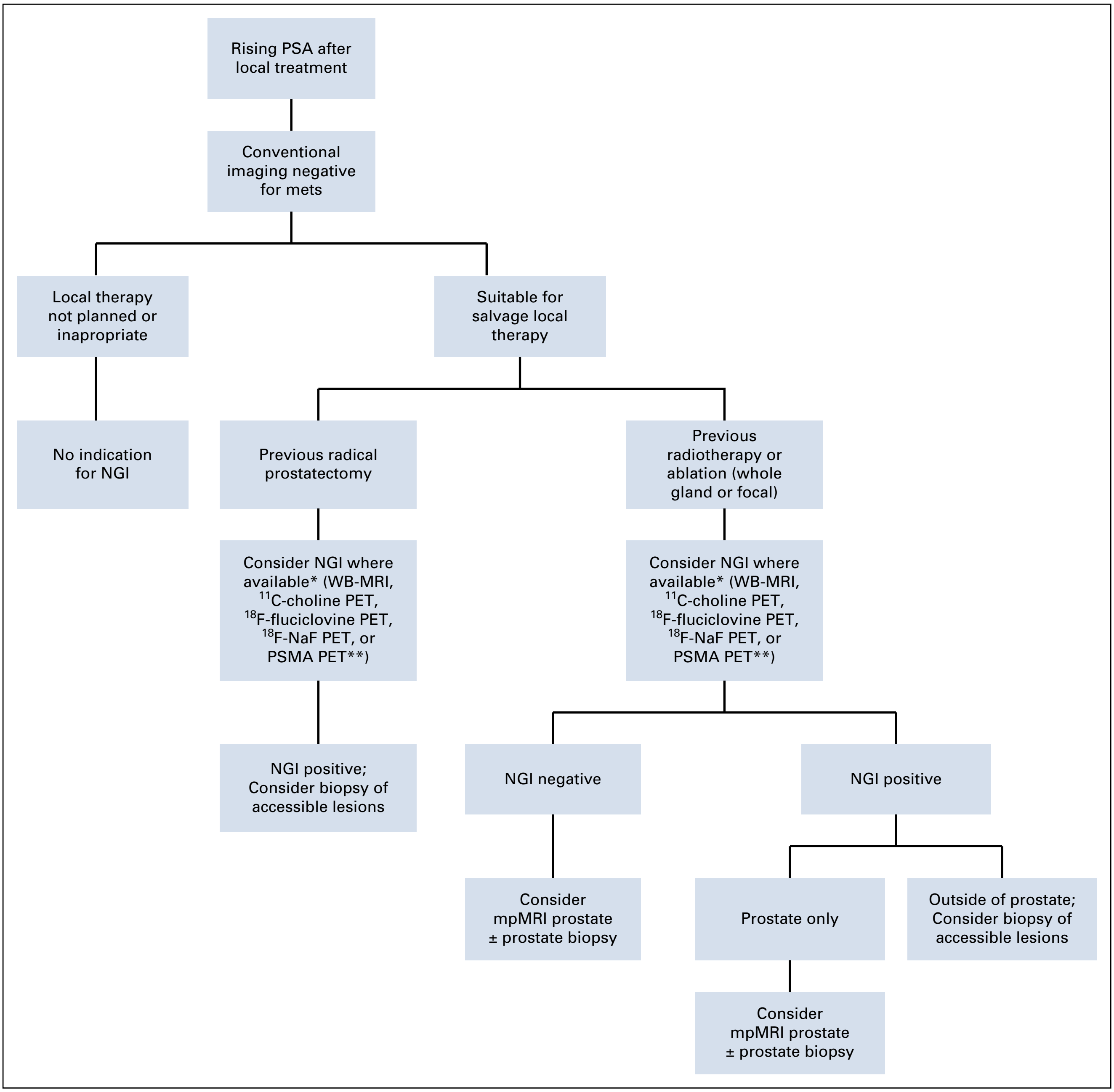

FIG 2. Imaging algorithm for patients with rising prostate-specific antigen (PSA) after local treatment. (*) For men for whom salvage local therapy (e.g. salvage radiation, salvage prostatectomy) is an option, there is evidence supporting the use of NGI to assess local or distant sites of disease, which may guide therapy away from salvage local therapy if indicative of distant metastatic disease. (**) There is enthusiasm for the potential added value of PSMA PET/CT and PET/MRI for the assessment of the local and metastatic extent of prostate cancer in this context, although PSMA imaging is not currently FDA approved and should thus be only performed as part of a clinical trial or other controlled research setting. Mets, metastatic disease; mpMRI, multiparametric magnetic resonance imaging; NGI, next-generation imaging; PET, positron emission tomography; PSMA, prostate-specific membrane antigen; WB, whole body.

conventional imaging become reclassified as having metastatic disease after NGI. This may alter treatment decisions, with unknown consequences on the overall disease course.[13] The Will Rogers phenomenon also makes comparison with historical standards and controls difficult, with the risk of artifactual changes in stage-specific survival.[14] The guideline summary addresses the goals of imaging in advanced prostate cancer, considering conventional imaging techniques, newer NGI modalities, as well as unmet needs, the potential impact of imaging according to different advanced prostate cancer clinical disease states, and the type of imaging that is most appropriate in each scenario. This guideline focuses on appropriate utility of imaging for advanced prostate cancer and not on treatment decisions; any discussion of treatment decisions is given in the context of the impact of imaging on clinical decision making. Evaluation of treatment options and treatment decision is beyond the scope of this guideline. Additional information is available at www.asco.org/genitourinary-cancer-guidelines. Patient information is available at www.cancer.net.

WHAT IS PRACTICE CHANGING

NGI, such as prostate-targeted PET, including PET/CT and PET/MRI, as well as whole-body MRI, has generated considerable excitement and interest for the accurate staging of patients with advanced prostate cancer, as well as to follow response to therapy. This guideline examines their role for different disease states of advanced prostate cancer, including newly diagnosed clinical high-risk disease, suspected or confirmed metastatic disease, recurrent disease, or progressive disease while under treatment. Using this disease states model, the role of NGI was examined for several common clinical scenarios of prostate cancer.

- For men with clinical high-risk disease at initial diagnosis, the literature supports addition of NGI when conventional imaging (CT, radionuclide bone scan, and/or multiparametric prostate MRI) is equivocal or suspicious for metastatic disease.
- For men with biochemical evidence of recurrent disease after local therapy, who have negative conventional imaging, for whom salvage therapy is planned, NGI is indicated to assess the presence of local or distant site and burden of disease, as this may alter treatment planning.
- For men with demonstrable metastatic disease at initial presentation on conventional imaging, NGI may clarify the burden of disease and alter treatment plans.
- For men with nonmetastatic castrate-resistant prostate cancer (CRPC), NGI can be considered to assess for unrecognized metastatic disease, although the clinical impact on therapy is limited.
- For men with metastatic CRPC with biochemical progression only, use of NGI is unclear and should be individualized, preferably as part of a clinical trial.

See Figure 1 for the imaging algorithm for high/very high-risk disease at initial presentation (per National Comprehensive Cancer Network). See Figure 2 for the imaging algorithm for patients with increasing PSA after local treatment.

AFFILIATIONS

[1]Sidney Kimmel Medical College at Thomas Jefferson University, Philadelphia, PA
[2]American Society of Clinical Oncology, Alexandria, VA
[3]Memorial Sloan Kettering Cancer Center, New York, NY

CORRESPONDING AUTHOR

American Society of Clinical Oncology, 2318 Mill Road, Suite 800, Alexandria, VA 22314; e-mail: guidelines@asco.org.

AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST AND DATA AVAILABILITY STATEMENT

Disclosures provided by the authors and data availability statement (if applicable) are available with this article at DOI https://doi.org/10.1200/OP.19.00699.

AUTHOR CONTRIBUTIONS

Conception and design: All authors
Administrative support: All authors
Collection and assembly of data: All authors
Data analysis and interpretation: All authors
Manuscript writing: All authors
Final approval of manuscript: All authors
Accountable for all aspects of the work: All authors

ACKNOWLEDGMENT

Optimum Imaging Strategies for Advanced Prostate Cancer: ASCO Guideline was developed and written by: Edouard J. Trabulsi, R. Bryan Rumble, Hossein Jadvar, Thomas Hope, Martin Pomper, Baris Turkbey, Andrew B. Rosenkrantz, Sadhna Verma, Daniel J. Margolis, Adam Froemming, Aytekin Oto, Andrei Purysko, Matthew I. Milowsky, Heinz-Peter Schlemmer, Matthias Eiber, Michael J. Morris, Peter L. Choyke, Anwar Padhani, Jorge Oldan, Stefano Fanti, Suneil Jain, Peter A. Pinto, Kirk A. Keegan, Christopher R. Porter, Jonathan A. Coleman, Glenn S. Bauman, Ashesh B. Jani, Jeffrey M. Kamradt, Westley Sholes, and Hebert Alberto Vargas.

REFERENCES

1. Sanda MG, Cadeddu JA, Kirkby E, et al: Clinically localized prostate cancer: AUA/ASTRO/SUO guideline. Part I: Risk stratification, shared decision making, and care options. J Urol 199:683-690, 2018
2. Siegel RL, Miller KD, Jemal A: Cancer statistics, 2019. CA Cancer J Clin 69:7-34, 2019
3. Ryan CJ, Shah S, Efstathiou E, et al: Phase II study of abiraterone acetate plus prednisone in chemotherapy-naïve metastatic castration-resistant prostate cancer demonstrating radiographic flare discordant with serologic measures of response. Clin Cancer Res 17:4854-4861, 2011
4. Scher HI, Halabi S, Tannock I, et al: Design and end points of clinical trials for patients with progressive prostate cancer and castrate levels of testosterone: Recommendations of the Prostate Cancer Clinical Trials Working Group. J Clin Oncol 26:1148-1159, 2008
5. Scher HI, Morris MJ, Stadler WM, et al: Trial design and objectives for castration-resistant prostate cancer: Updated recommendations from the Prostate Cancer Clinical Trials Working Group 3. J Clin Oncol 34:1402-1418, 2016
6. Kluetz PG, Pierce W, Maher VE, et al: Radium Ra 223 dichloride injection: U.S. Food and Drug Administration drug approval summary. Clin Cancer Res, 20: 9-14, 2014

7. Morris MJ, Molina A, Small EJ, et al: Radiographic progression-free survival as a response biomarker in metastatic castration-resistant prostate cancer: COU-AA-302 results. J Clin Oncol 33:1356-1363, 2015
8. Rathkopf DE, Beer TM, Loriot Y, et al: Radiographic progression-free survival as a clinically meaningful end point in metastatic castration-resistant prostate cancer: The PREVAIL randomized clinical trial. JAMA Oncol 4:694-701, 2018
9. Imbriaco M, Larson SM, Yeung HW, et al: A new parameter for measuring metastatic bone involvement by prostate cancer: the Bone Scan Index. Clin Cancer Res 4:1765-1772, 1998
10. Anand A, Morris MJ, Kaboteh R, et al: Analytic validation of the automated bone scan index as an imaging biomarker to standardize quantitative changes in bone scans of patients with metastatic prostate cancer. J Nucl Med 57:41-45, 2016
11. Armstrong AJ, Anand A, Edenbrandt L, et al: Phase 3 assessment of the automated bone scan index as a prognostic imaging biomarker of overall survival in men with metastatic castration-resistant prostate cancer: A secondary analysis of a randomized clinical trial. JAMA Oncol 4:944-951, 2018
12. Bach-Gansmo T, Nanni C, Nieh PT, et al: Multisite experience of the safety, detection rate and diagnostic performance of fluciclovine (^{18}F) positron emission tomography/computerized tomography imaging in the staging of biochemically recurrent prostate cancer. J Urol 197:676-683, 2017
13. Vapiwala N, Hofman MS, Murphy DG, et al: Strategies for evaluation of novel imaging in prostate cancer: Putting the horse back before the cart. J Clin Oncol 37: 765-769, 2019
14. Christensen D: The Will Rogers phenomenon: Roping the effects of a new cancer staging system. J Natl Cancer Inst 95:1105-1106, 2003
15. Roach M III, Hanks G, Thames H, Jr., et al: Defining biochemical failure following radiotherapy with or without hormonal therapy in men with clinically localized prostate cancer: Recommendations of the RTOG-ASTRO Phoenix Consensus Conference. Int J Radiat Oncol Biol Phys 65:965-974, 2006
16. Tilki D, Kim SI, Hu B, et al: Ultrasensitive prostate specific antigen and its role after radical prostatectomy: A systematic review. J Urol 193:1525-1531, 2015

AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

Optimum Imaging Strategies for Advanced Prostate Cancer: ASCO Guideline Summary

The following represents disclosure information provided by authors of this manuscript. All relationships are considered compensated unless otherwise noted. Relationships are self-held unless noted. I = Immediate Family Member, Inst = My Institution. Relationships may not relate to the subject matter of this manuscript. For more information about ASCO's conflict of interest policy, please refer to www.asco.org/rwc or ascopubs.org/op/site/ifc/journal-policies.html.

Open Payments is a public database containing information reported by companies about payments made to US-licensed physicians (Open Payments).

Edouard J. Trabulsi
Consulting or Advisory Role: GenomeDx
Speakers' Bureau: Johnson & Johnson, Janssen, Astellas Medivation, Pfizer

R. Bryan Rumble
Employment: Park Lane Terrace (I)

No other potential conflicts of interest were reported.